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資料結構
Data Structure

Lab 08

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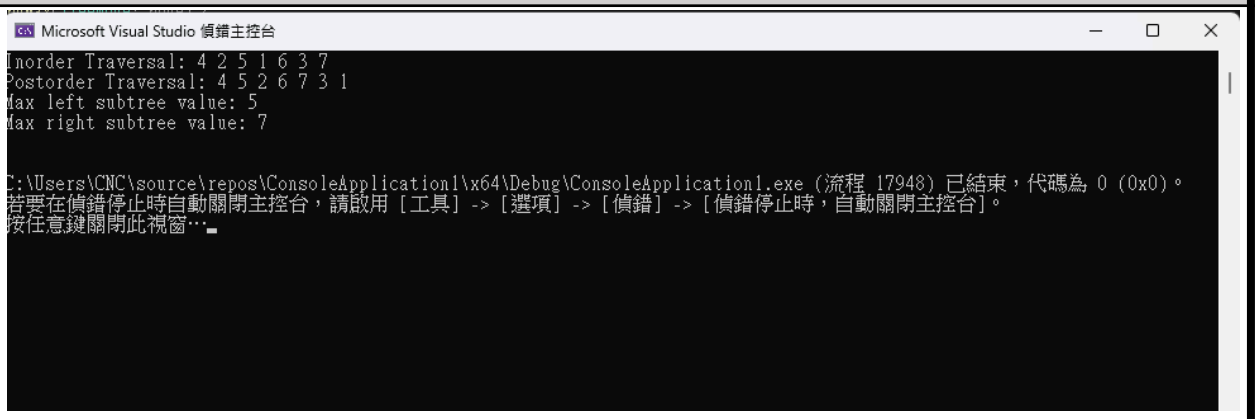
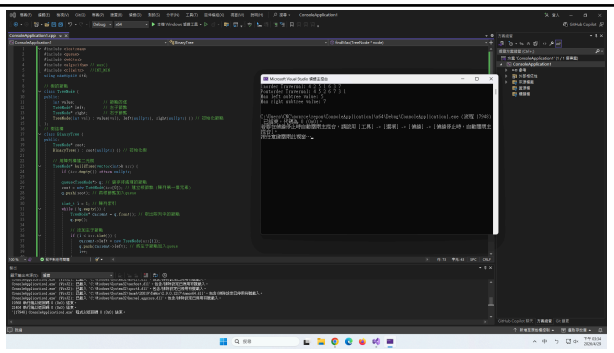
Lab08-Ex1. (題目編號)
Exercise: Implement Postorder Traversal
Code

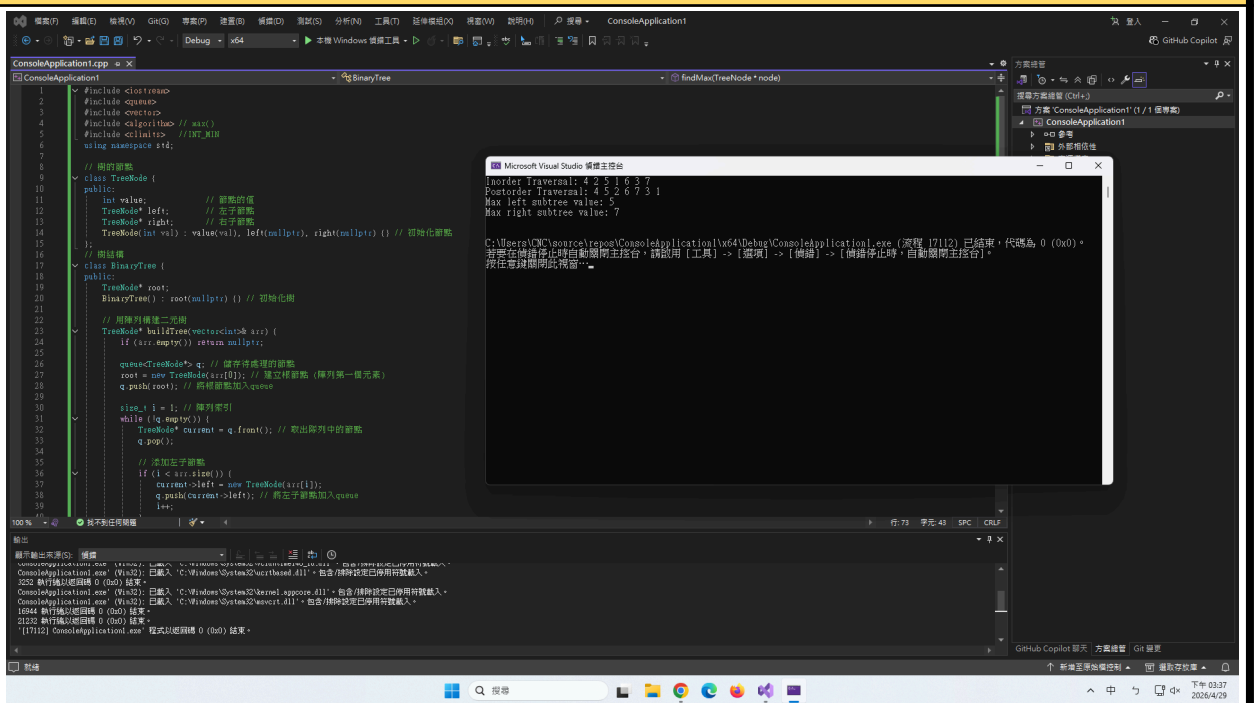
```
#include <iostream>
#include <queue>
#include <vector>
#include <algorithm> // max()
#include <climits>   //INT_MIN
using namespace std;

// 樹的節點
class TreeNode {
public:
   int value;          // 節點的值
   TreeNode* left;      // 左子節點
   TreeNode* right;      // 右子節點
   TreeNode(int val) : value(val), left(nullptr), right(nullptr) {} // 初始化節點
};
// 樹結構
class BinaryTree {
public:
   TreeNode* root;
   BinaryTree() : root(nullptr) {} // 初始化樹

   // 用陣列構建二元樹
   TreeNode* buildTree(vector<int>& arr) {
      if (arr.empty()) return nullptr;

      queue<TreeNode*> q; // 儲存待處理的節點
      root = new TreeNode(arr[0]); // 建立根節點 (陣列第一個元素)
      q.push(root); // 將根節點加入queue

      size_t i = 1; // 陣列索引
      while (!q.empty()) {
         TreeNode* current = q.front(); // 取出隊列中的節點
         q.pop();

         // 添加左子節點
         if (i < arr.size()) {
            current->left = new TreeNode(arr[i]);
            q.push(current->left); // 將左子節點加入queue
            i++;
         }
```

```
        // 添加右子節點
        if (i < arr.size()) {
            current->right = new TreeNode(arr[i]);
            q.push(current->right); // 將右子節點加入queue
            i++;
        }
    }

    return root;
}
// 中序遍歷
void inorderTraversal(TreeNode* node) {
    if (node == nullptr) return; // 如果節點為空，忽略

    inorderTraversal(node->left);  // 遍歷左子樹
    cout << node->value << " ";   // 訪問當前節點
    inorderTraversal(node->right); // 遍歷右子樹
}

// 後序遍歷
void postorderTraversal(TreeNode* node) {
    if (node == nullptr) return;

    postorderTraversal(node->left);  // 1. 先走左邊
    postorderTraversal(node->right); // 2. 再走右邊
    cout << node->value << " ";     // 3. 最後才印出自己
}

int findMax(TreeNode* node) {
    if (node == nullptr) return INT_MIN; // 如果是空的，回傳整數最小值

    int maxVal = node->value;
    int leftMax = findMax(node->left);
    int rightMax = findMax(node->right);

    return max(maxVal, max(leftMax, rightMax));
}

};
int main() {
    BinaryTree tree; // 宣告二元樹

    // 輸入陣列用於構建樹，NULL 表示空子節點
```

```
    vector<int> arr = { 1, 2, 3, 4, 5, 6, 7 };

    tree.buildTree(arr); // 建立樹

    // 中序遍歷輸出
    cout << "Inorder Traversal: ";
    tree.inorderTraversal(tree.root);
    cout << endl;

    // 後序遍歷輸出
    cout << "Postorder Traversal: ";
    tree.postorderTraversal(tree.root);
    cout << endl;
    if (tree.root != nullptr) {
        cout << "Max left subtree value: " << tree.findMax(tree.root->left) << endl;
        cout << "Max right subtree value: " << tree.findMax(tree.root->right) << endl;
    }
    cout << endl;

    return 0;
}
```







Lab01-Q1. (題目編號)

Code

Result
Discussion

Lab01-Q#. (題目編號)
Briefly describe the problem. (題目簡述)
Code

Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)

result		
(a)	(b)	(c)
	Show your result in tables or screenshots. (請使用表格或是截圖呈現結果。)	

Discussion
Discuss and conclude your results, or answer questions. (實驗討論、結論或回答問題。)